

Normalization Case Study: Community Arts Fair Catering Orders

Course: CSE 2221 - Database Management Systems
Outcome: CO3 | **Submission:** Printed hard copy by 24 November 2025
Scenario Owner: Lakeview Community Arts Fair (unique topic selected)

1. Scenario Overview

Student clubs submit a single catering form that lists event logistics plus multiple dishes. The form stores every dish (name, category, price) inside one cell, which introduces redundancy, anomalies, and mixed dependencies. The goal is to design the original unnormalized structure and normalize it to 3NF while showing every step with labeled artifacts.

2. Unnormalized Table (UNF) — Table 1

Table Name: FestivalCateringOrder_UNF (7 attributes, contains repeating group)

OrderCode	EventDate	OrganizerEmail	EventName	ValueZone	ZoneSupervisor	BackPage	MenuItemDetails
ORD-101	2025-03-12	eco.club@calnorth.edu	North Quad	555-8144	Silver		Veg Wrap Starter 5 ; Solar Salad Main 11 ; Citrus Cake Dessert 7
ORD-102	2025-03-15	dance.circle@calcampus.edu	East Lawn	555-2279	Gold		Garden Rolls Starter 6 ; Fire Grill Main 13
ORD-103	2025-03-19	robotics.hub@calnorth.edu	North Quad	555-8144	Gold		Veg Wrap Starter 5 ; Kinetic Curry Main 12 ; Circuit Sun-dae Dessert 8

OrderCode	EventDate	OrganizerEmail	VenueZone	ZoneSupervisorPhone	PackageTier	MenuItemDetails
ORD-104	2025-03-19	drama.club@scampus.edu	South Lawn	555-9902	Silver	Lantern Soup Starter 4 ; Stage Stew Main 10 ; Finale Tart Dessert 6
ORD-105	2025-03-22	chess.guild@scampus.edu	East Lawn	555-2279	Bronze	Garden Rolls Starter 6

Issues in Table 1 - Repeating group (MenuItemDetails) violates 1NF.

- Partial dependency: EventDate, OrganizerEmail, VenueZone, PackageTier rely on OrderCode only.

- Transitive dependency: VenueZone -> ZoneSupervisorPhone.

3. Functional Dependencies in UNF

- **FD1:** OrderCode -> EventDate, OrganizerEmail, VenueZone, PackageTier
- **FD2:** VenueZone -> ZoneSupervisorPhone
- **FD3:** DishName -> DishCategory, UnitPrice

4. Normalization Steps

Step 1: Convert to First Normal Form (1NF)

- **Action:** Flatten repeating menu data into individual rows.
- **New Table (Table 2):** FestivalOrderLine_1NF

OrderCode	DishName	EventDate	OrganizerEmail	VenueZone	ZoneSupervisorPhone	PackageTier	DishCategory	UnitPrice
ORD-101	Veg Wrap	2025-03-12	eco.club@scampus.edu	South Quad	555-8144	Silver	Starter	5
ORD-101	Solar Salad	2025-03-12	eco.club@scampus.edu	South Quad	555-8144	Silver	Main	11
ORD-102	Garden Rolls	2025-03-15	dance.circles@scampus.edu	East Lawn	555-2279	Gold	Starter	6

OrderCode	DishName	EventDate	OrganizerEmail	VenueZone	ZoneSupervisorPhone	PackageTier	DishCategory	UnitPrice
ORD-103	Circuit Sun-dae	2025-03-19	robotics.ln@champs.edu	North Quad	555-8144	Gold	Dessert	8

Primary Key: (OrderCode, DishName)

Remaining Dependencies:

- OrderCode -> EventDate, OrganizerEmail, VenueZone, PackageTier (partial)
- VenueZone -> ZoneSupervisorPhone (transitive)
- DishName -> DishCategory, UnitPrice (repetition of item info across orders)

Step 2: Remove Partial Dependencies (Reach 2NF)

Split order header facts from dish details.

- **Table 3:** FestivalOrderHeader(OrderCode PK, EventDate, OrganizerEmail, VenueZone, ZoneSupervisorPhone, PackageTier)
- **Table 4:** FestivalOrderDish(OrderCode PK/FK, DishName PK, DishCategory, UnitPrice)

Status:

- Header table no longer repeats per dish.
- Dish table still repeats DishCategory and UnitPrice for each order.
- VenueZone -> ZoneSupervisorPhone still transitive in Table 3.

Step 3: Remove Transitive Dependencies (Reach 3NF)

Introduce lookup tables for zones and menu catalog, and create a pure association table.

- **Table 5:** VenueZoneDirectory(<u>VenueZone</u>, ZoneSupervisorPhone)
- **Table 6:** MenuCatalog(<u>DishName</u>, DishCategory, UnitPrice)
- **Table 7:** FestivalOrderHeader(<u>OrderCode</u>, EventDate, OrganizerEmail, VenueZone*, PackageTier)
- **Table 8:** FestivalOrderItem(<u>OrderCode</u>*, <u>DishName</u>*)

Resulting Dependencies:

- OrderCode -> EventDate, OrganizerEmail, VenueZone, PackageTier (Table 7).
- VenueZone -> ZoneSupervisorPhone (Table 5).
- DishName -> DishCategory, UnitPrice (Table 6).

- **OrderCode**, **DishName** identifies each ordered dish in Table 8 with no leftover partial/transitive dependencies.

5. Final Schema Summary (PK underlined, FK with *)

1. VenueZoneDirectory(VenueZone, ZoneSupervisorPhone)
2. MenuCatalog(DishName, DishCategory, UnitPrice)
3. FestivalOrderHeader(OrderCode, EventDate, OrganizerEmail, VenueZone*, PackageTier)
4. FestivalOrderItem(OrderCode*, DishName*)

All primary and foreign keys are now clearly defined, satisfying the assignment requirements.

6. Impact of Normalization

Normalization separated repeating menu entries, isolated zone contact details, and centralized dish pricing. Updates now touch a single row (e.g., change a supervisor's phone once in **VenueZoneDirectory**), insertions no longer require null placeholders, and deletions cannot accidentally remove shared information. The 3NF schema therefore eliminates redundancy-driven anomalies and enforces integrity through explicit relationships.

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